```
scale=10; 1/3
.3333333333

scale=20; 1/3
.33333333333333333333

scale=30; 1/3
.333333333333333333333333333333
```

Figure 1: bc in action



Figure 2: A simple calculation in Scilab

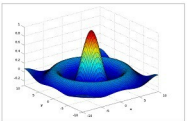


Figure 3: Output from GNU Octave

Scilab clearly outperforms MATLAB, and that is, price! You absolutely don't have to pay anything to use Scilab, whereas MATLAB is a relatively costly scientific tool. Scilab can be used as a programming environment and an interactive mathematical shell. Figure 2 shows how Scilab works as an interactive mathematical shell.

GNU Octave

GNU Octave is another important mathematical tool primarily used for numerical computations. It is part of the GNU Project and is offered as free and open source software licensed under the GNU General Public License. GNU Octave is one of the two very powerful open source alternatives to MATLAB, the other being Scilab. But GNU Octave gives far more importance to syntactic compatibility with MATLAB than Scilab does. To prove our point, we have copied and executed the following MATLAB code from the Wikipedia article on MATLAB, on both Scilab and GNU Octave.

```
[X,Y] = meshgrid(-10:0.25:10,-10:0.25:10);
f = sinc(sqrt((X/pi).^2+(Y/pi).^2));
surf(X,Y,f);
axis([-10 10 -10 10 -0.3 1])
xlabel('{\bfx}')
ylabel('{\bfy}')
zlabel('{\bfsinc} ({\bfR})')
```

Scilab failed to produce an output whereas GNU Octave successfully executed the program. The output obtained from GNU Octave is the same as the output obtained from MATLAB. The output is shown in Figure 3, and you can compare it with the figure given in the Wikipedia article on MATLAB to fully comprehend the syntactic similarity of GNU Octave and MATLAB. GNU Octave can also be used as a programming environment and an interactive shell.

Maxima

Maxima is a computer algebra system that is widely used in the fields of mathematics like algebra, calculus, etc. Maxima is developed using Lisp, and it is a cross-platform tool that runs on UNIX, Linux, Windows, Android, MacOS, etc. It is free and open source software licensed under the GNU General Public License. Due to the complexity involved in developing computer algebra systems, there are only a few powerful tools like Maxima and SageMath. A very popular graphical user interface (GUI) for Maxima is called wxMaxima, and it can be used as a programming environment and an interactive shell. Figure 4 shows how Maxima can be used to perform integration. I have used a simple function to perform integration, but you can also find the integral of complex functions using Maxima.

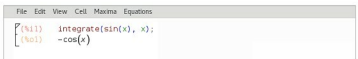


Figure 4: Integration using Maxima

SageMath

SageMath is a computer algebra system that has widespread use in various fields of mathematics like algebra, combinatorics, graph theory, numerical analysis, number theory, calculus, statistics, etc. Computer algebra systems are mostly used for symbolic computing, rather than numeric computing and number crunching. The large number of packages necessary for a powerful computer algebra system makes them very rare and due to this reason, SageMath is a very important tool. It is a free and open source tool licensed under GNU General Public License, and is a competitor to many other mathematical tools like Maple, Mathematicia, MATLAB, etc, which are proprietary software. SageMath is not only competent enough to replace any of these tools but also absolutely free. It can be used in two modes, as a programming environment and an interactive shell. The browser-based interactive shell called Notebook has the ability to remember previous inputs and outputs for review and reuse. SageMath is a relatively new tool with most of the development being done using Python; the initial release came out in 2005. Figure 5 shows the logo of SageMath.



Figure 5: The logo of SageMath

R

R is a mathematical tool mainly used for statistical computing. It is used by mathematicians in general and by statisticians in particular for data mining and developing statistical software. The development of R is supervised by the R Foundation for Statistical Computing. R is a cross-platform tool that can be used in various operating systems like Linux, Microsoft Windows, etc. R is free and open source